

Quantum-enhanced Computer Vision: Going Beyond Classical Algorithms

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Quantum-enhanced Computer Vision: Going Beyond Classical Algorithms

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Abstract—Quantum-enhanced Computer Vision (QeCV) is a new research field at the intersection of computer vision, optimisation theory, machine learning and quantum computing. It has high potential to transform how visual signals are processed and interpreted with the help of quantum computing that leverages quantum-mechanical effects in computations inaccessible to classical (i.e. non-quantum) computers. In scenarios where existing non-quantum methods cannot find a solution in a reasonable time or compute only approximate solutions, quantum computers can provide, among others, advantages in terms of better time scalability for multiple problem classes. Parametrised quantum circuits can also become, in the long term, a considerable alternative to classical neural networks in computer vision. However, specialised and fundamentally new algorithms must be developed to enable compatibility with quantum hardware and unveil the potential of quantum computational paradigms in computer vision.

This survey contributes to the existing literature on QeCV with a holistic review of this research field. It is designed as a quantum computing reference for the computer vision community, targeting computer vision students, scientists and readers with related backgrounds who want to familiarise themselves with QeCV. We provide a comprehensive introduction to QeCV, its specifics, and methodologies for formulations compatible with quantum hardware and QeCV methods, leveraging two main quantum computational paradigms, i.e. gate-based quantum computing and quantum annealing. We elaborate on the operational principles of quantum computers and the available tools to access, program and simulate them in the context of QeCV. Finally, we review existing quantum computing tools and learning materials and discuss aspects related to publishing and reviewing QeCV papers, open challenges and potential social implications.

Index Terms—Quantum-enhanced Computer Vision, Applied Quantum Computing, Quantum Algorithms.

cs.CV] 8 Oct 2025

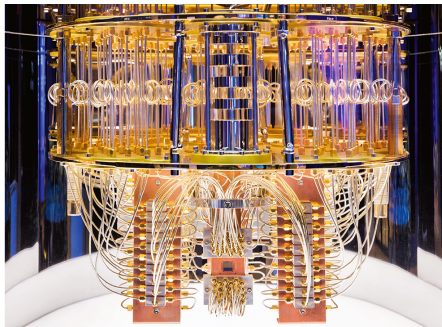
Agenda

- 1 Background
 - Motivation and Goals
- 2 Operational Principles
 - Basics and Paradigms
- 3 Methods
 - The Truth, the Hype and the Tools
- 4 Advanced Horizons
 - Discrete, Continuous, Probabilistic
- 5 Open Challenges
 - Encoding, Training, Scaling

Background

Motivation and Goals

Why Quantum Computing?



Expand computing hardware
with an entirely new device type

- CPU/GPU/**QPU**/...
- Compute with laws of **quantum mechanics**.
- **Any computational advantages beyond classical hardware?**

Why Quantum Computing?

QUBIT
QUBIT
QUBIT

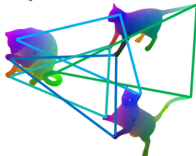
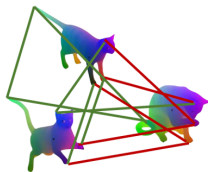
Birdal and Golyanik,
2020



Seelbach Benkner,
2021



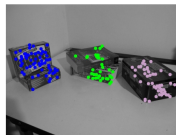
Li, 2020



Birdal, 2021



Bhatia, 2023



Farina, 2023

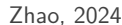
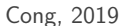


Kuete, 2025

And many more...

**Optimization
& Energy minimization**

- Provable quantum speedups exist for specific problems (e.g., Shor/Grover/...).
- **Can similar advantages emerge in computer vision?**

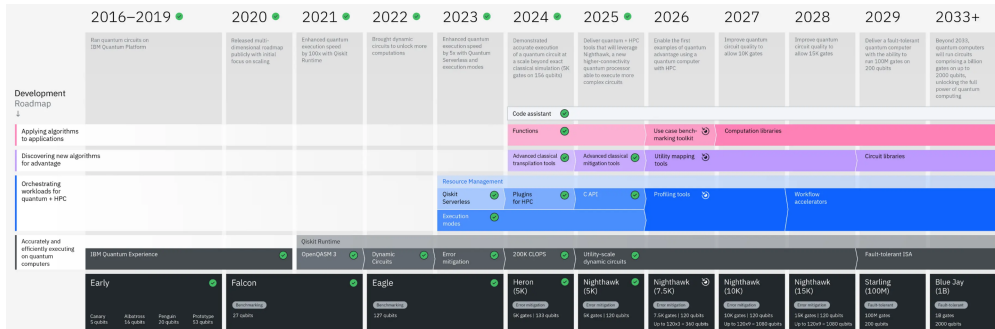


Quantum Machine Learning

- We can essentially use and train quantum computers like neural networks.
- **Can quantum actually outperform classical neural networks?**

Quantum Computing Roadmaps

IBM Road Map



And several companies in the race (physical/logical qubit support):

	Year 2026	~ Year 2032
IBM (Gate model) https://www.ibm.com/roadmaps/quantum/	360+/-	- / 2,000+
Google (Gate-based) https://quantumai.google/roadmap	100+/-	1M / 10,000+
IonQ (Gate model) https://www.ionq.com/roadmap	256+ / 64+	2M / 80,000+
D-Wave (Gate model) https://www.dwavequantum.com/media/ciwepakh/gate-model-roadmap.jpg	17/-	- / 100+
D-Wave (Annealer) https://www.dwavequantum.com/solutions-and-products/product-overview/	4,400+/-	- / -

IBM roadmap: <https://www.ibm.com/quantum/hardware>

Operational Principles

Basics and Paradigms

Quantum bit

Any particle that has two basis states and can be in a superposition of them.

Qubit

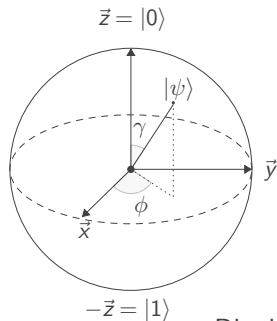
Let $|0\rangle$ and $|1\rangle$ be two basis vectors of $\mathbb{C}^2$, with

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

A **qubit** is a one-particle system whose state vector is expressed as

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle,$$

with $\alpha, \beta \in \mathbb{C}$ subject to $|\alpha|^2 + |\beta|^2 = 1$.



Bloch sphere

Multi-qubit systems

The state of a n -qubit system is given by the **tensor (Kronecker)** product

$$|\psi\rangle = |\psi_1\rangle \otimes \cdots \otimes |\psi_n\rangle \in \mathbb{C}^{2^n}$$

of the n single-qubit states.

Basis states are correspondingly formed by tensor products of $|0\rangle$ and $|1\rangle$.

Example for $n = 3$:

$$|\psi\rangle = \alpha_{000} |000\rangle + \alpha_{001} |001\rangle + \cdots + \alpha_{111} |111\rangle$$

$\uparrow$
 $|0\rangle \otimes |0\rangle \otimes |0\rangle$
or
 $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix}$
or 0 in binary basis

$\uparrow$
 $|0\rangle \otimes |0\rangle \otimes |1\rangle$
or
 $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix}$
or 1 in binary basis

$\uparrow$
 $|1\rangle \otimes |1\rangle \otimes |1\rangle$
or
 $\begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix}$
or 7 in binary basis

Entanglement

Not every multi-qubit states can be written as a tensor product of single-qubit states.

So-called **entangled states** cannot be written in this form.

As an example, for the state

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad (1)$$

there exists no $|\psi_1\rangle = \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix}$ and $|\psi_2\rangle = \begin{pmatrix} \alpha_2 \\ \beta_2 \end{pmatrix}$ such that

$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle = \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} \otimes \begin{pmatrix} \alpha_2 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} \alpha_1\alpha_2 \\ \alpha_1\beta_2 \\ \beta_1\alpha_2 \\ \beta_1\beta_2 \end{pmatrix}. \quad (2)$$

Measurement in a nutshell the Bloch sphere

Recall

A qubit's state vector

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1$$

collapses to the state $|0\rangle$ with probability $|\alpha|^2$ and to the state $|1\rangle$ with probability $|\beta|^2$.

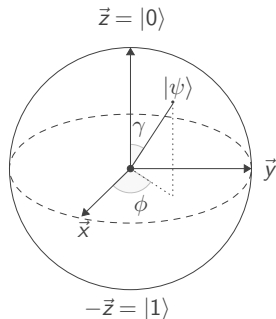
These are merely the results of

$$|\alpha|^2 = \langle\psi|0\rangle\langle 0|\psi\rangle \quad \text{and} \quad |\beta|^2 = \langle\psi|1\rangle\langle 1|\psi\rangle.$$

If we collect $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$ in a matrix

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

Then $\langle\psi|Z|\psi\rangle = |\alpha|^2 - |\beta|^2$ is the z-coordinate of $|\psi\rangle$ in $\mathbb{R}^3$.



Adiabatic vs. Gate-based QC

At any time $t \in [0, T]$, the evolution of the state vector $|\psi(t)\rangle$ of a quantum system obeys Schrödinger's equation

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = H(t) |\psi(t)\rangle ,$$

where H is a **Hermitian** operator known as the system-driven **Hamiltonian**.

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = H(t) |\psi(t)\rangle$$

$$H(t) = \mathcal{A}(t)H_I + \mathcal{B}(t)H_P$$

Adiabatic quantum computing (AQC)

H_I, H_P : initial and problem Hamiltonians

$\mathcal{A}(t), \mathcal{B}(t)$: scheduling functions

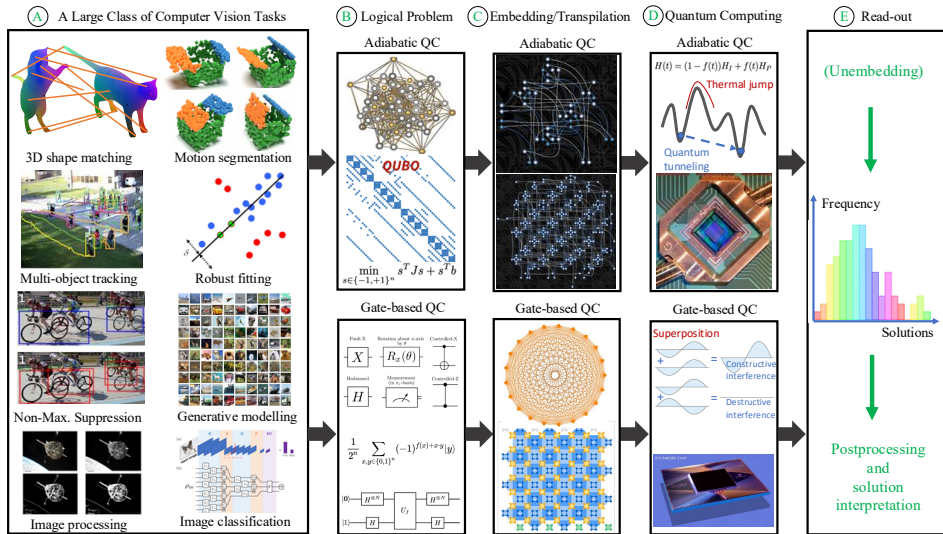
$$|\psi(t)\rangle = U(t) |\psi(0)\rangle$$

Gate-based quantum computing (GQC)

$U(t)$: unitary operator

$U(t)$: depends only on t

Two Paradigms: AQC vs. GQC

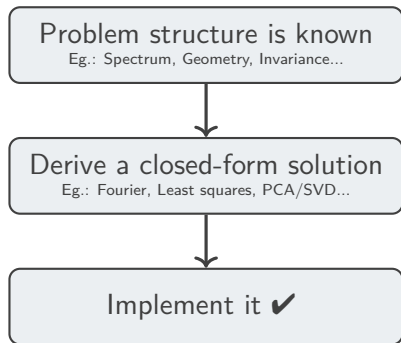


Methods

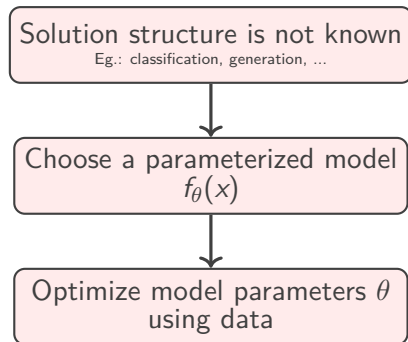
The Truth, the Hype and the Tools

Analytical vs. Learned Algorithms

Analytical



Learned / Variational



The same applies in quantum computing

Analytical AQC/GQC algorithms ↔ Variational AQC/GQC algorithms

What is AQC Good At?

QUBO: Quadratic Unconstrained Binary Optimization ¹

Given $J \in \mathbb{R}^{k \times k}$, $h \in \mathbb{R}^k$, minimize

$$\min_{s \in \{-1, 1\}^k} s^\top J s + h^\top s.$$

Problem Hamiltonian

Define a so-called **problem Hamiltonian**

$$H_P = \sum_{i,j=1}^k J_{ij} Z_i Z_j + \sum_{i=1}^k h_i Z_i,$$
$$Z_i = \dots \otimes I \otimes \underbrace{Z}_{\text{pos. } i} \otimes I \otimes \dots$$

For a state

$$|q\rangle = \bigotimes_{i=1}^k |q_i\rangle, \quad |q_i\rangle = \begin{cases} |0\rangle, & \text{if } s_i = 1 \\ |1\rangle, & \text{if } s_i = -1, \end{cases}$$

$$s^\top J s + h^\top s = \langle q | H_P | q \rangle$$

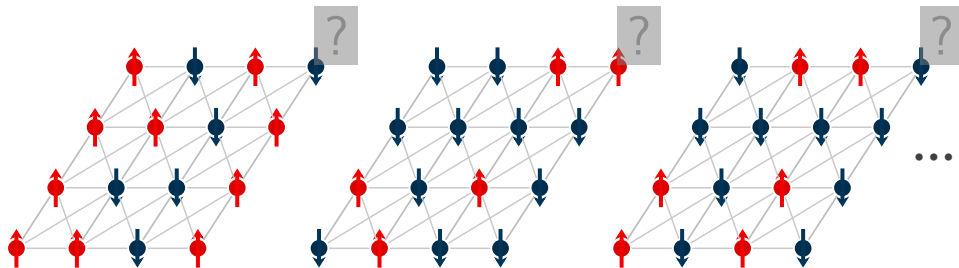
= energy of the quantum system in state $|q\rangle$.

What is AQC Good At?

So there is an energy associated with each qubit configuration with respect to a given Hamiltonian H_P .

The question becomes: **Which qubit configuration minimizes the energy?**

→ This would be the minimizer of the QUBO.



What is AQC Good At?

Adiabatic Quantum Computing ¹

A system evolving under the Schrödinger equation

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle$$

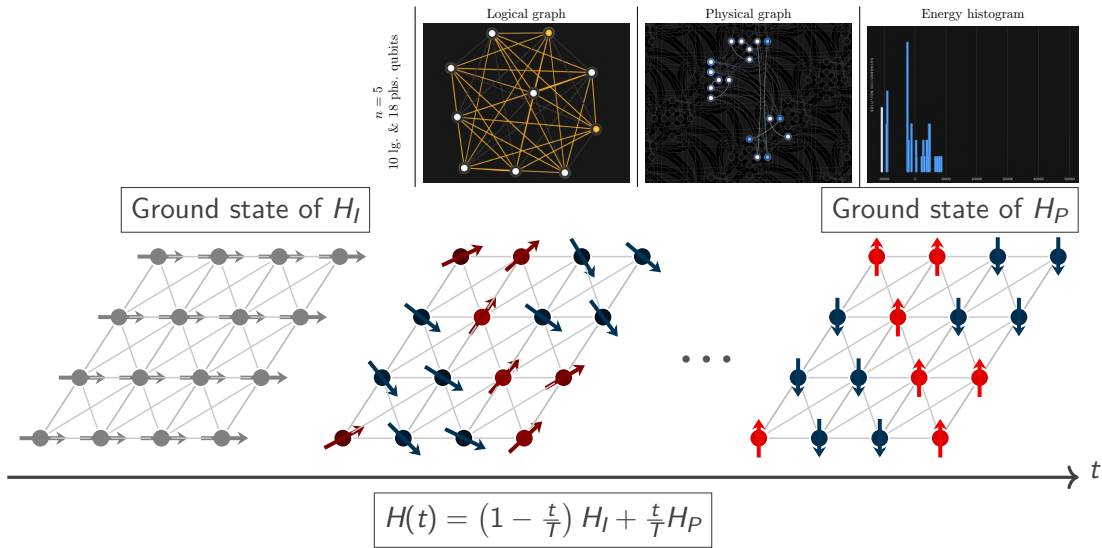
and **starting in the ground state of $H(0)$ remains in the ground state of $H(t)$** if the $H(t)$ changes slowly enough and the gap to the first excited state stays sufficiently large.

Define an initial H_I whose ground state is easy to prepare and let

$$H(t) = \left(1 - \frac{t}{T}\right) H_I + \frac{t}{T} H_P.$$

The adiabatic theorem brings the system in the ground state of H_P !!! ✓

What is AQC Good At?



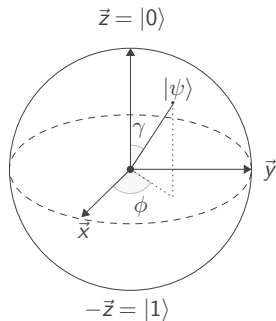
What is GQC Good At?

Computation yields rotating the qubits in the Bloch sphere:

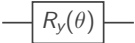
- Requires **unitary transformations** called gates.
- The initial state of the qubits will be assumed to be $|0\rangle$.
- On one qubit, a gate will just be a 2×2 unitary operator

$$U = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}.$$

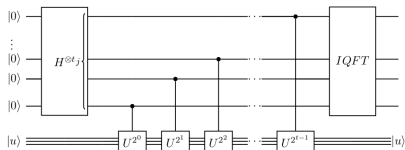
- Take the Kronecker product of single-qubit gates to form a gate on n qubits: $U = U_1 \otimes \cdots \otimes U_n$.
- Use controlled gates to form entangled states.



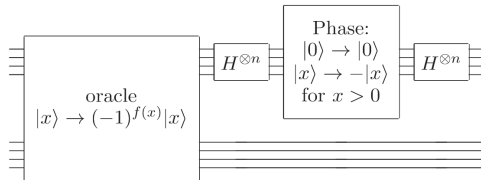
What is GQC Good At?

Gate name	Matrix form	Circuit	Notation
Pauli-X	$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$		$X 0\rangle = 1\rangle$
Hadamard	$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$		$H 0\rangle = \frac{1}{\sqrt{2}}(0\rangle + 1\rangle)$
Multi-qubit gate	$X \otimes H = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{pmatrix}$		$X \otimes H 00\rangle = \frac{1}{\sqrt{2}}(10\rangle + 11\rangle)$
Controlled-X on $ q_0q_1\rangle$	$I \otimes X^{q_0} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$		$I \otimes X^{q_0} 00\rangle = 00\rangle$ $I \otimes X^{q_0} 10\rangle = 11\rangle$
Rotation R_y	$R_y(2\theta) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$		$R_y(2\theta) 0\rangle = \cos(\theta) 0\rangle - \sin(\theta) 1\rangle$

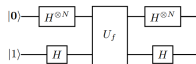
What is GQC Good At?



Phase estimation
Used in Shor's factoring
Shor 1997



Grover iteration
Used in search
Grover 1996



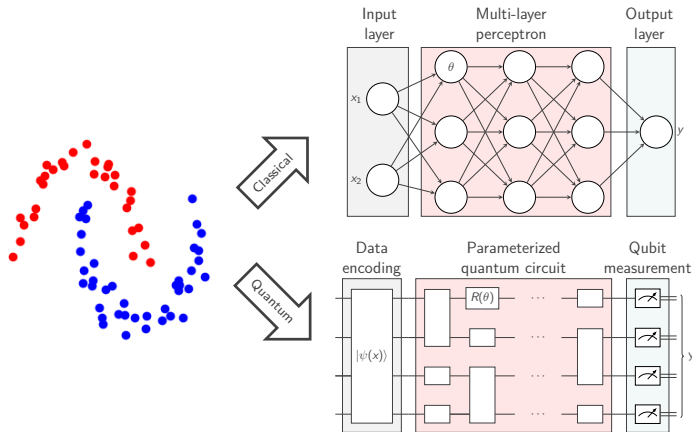
Bernstein-Vazirani
Used for Fourier sampling
Bernstein-Vazirani 1997

And many more...

What is GQC Good At?

What if we do not know the analytical circuit that solves our problem?

→ Train circuits like neural networks.



Parameter shift-rule¹:

$$\frac{\partial}{\partial \theta} f = \frac{1}{2} \left(f\left(\theta + \frac{\pi}{2}\right) - f\left(\theta - \frac{\pi}{2}\right) \right)$$

- Differentiable
- Trainable

¹ Schuld et al. Evaluating analytic gradients on quantum hardware. *Physical Review A*, 2019.

Advanced Horizons

Discrete, Continuous, Probabilistic

QUBOs and Discrete Optimization

What quantum computers are good at

$$\min_{s \in \{-1,1\}^k} s^T J s + h^T s$$

Substitution:
 $s = 2x - 1 \rightarrow$

$$\min_{x \in \{0,1\}^k} x^T Q x$$

Quadratic Assignment:

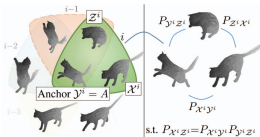
QGM [Seelbach Benkner et al, 2020]
Q-Match[Seelbach Benkner et al, 2021]
QuCOOP[Kuete Meli et al, 2025]



$$\min_{p \in \{0,1\}^{n^2} | Ap=b} p^T Q p$$

Pseudo Boolean Problems:

CCUANTUMM [Bhatia et al, 2023]
QuCOOP[Kuete Meli et al, 2025]



$$\min_{x \in \{0,1\}^n} \sum_{S \subseteq [n]} w_S \prod_{j \in S} x_j$$

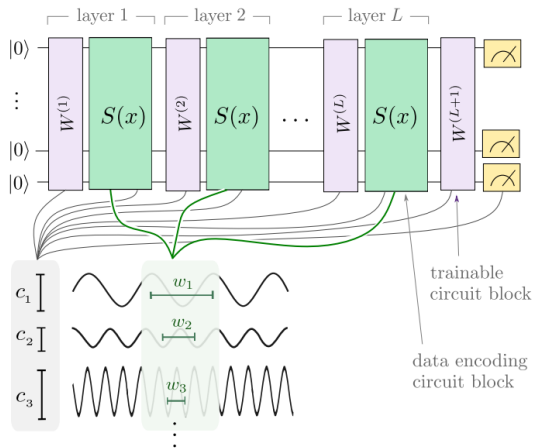
Probabilistic Sampling:

K-Means [Zaech et al, 2024]



$$\min_{Z \in \{0,1\}^n} -\log p(Z|X)$$
$$p(Z|X) \propto p(X, Z)$$

Quantum Implicit Representations



What quantum computers are good at

Data re-uploading realizes Fourier series:

$$f_{\theta}(x) = \langle 0 | U^{\dagger}(x, \theta) M U(x, \theta) | 0 \rangle$$

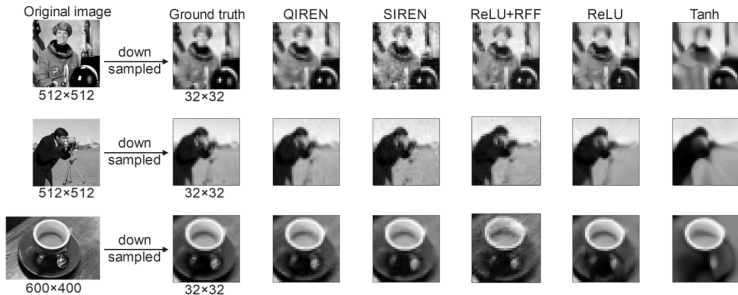
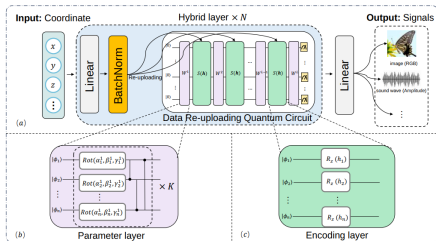
$$= \sum_{\omega \in \Omega} c_{\omega}(\theta) e^{i\omega x}.$$

We have $|\Omega| \leq 4^{nL}$, for

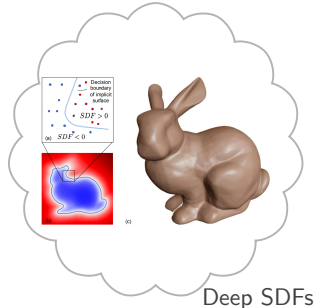
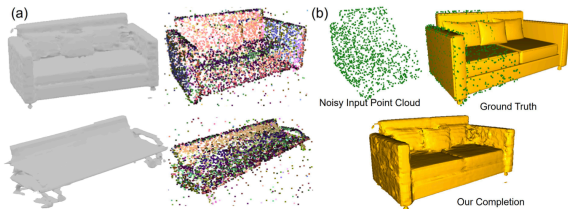
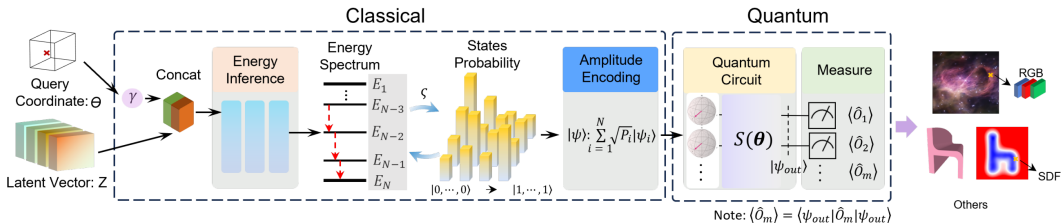
- n -qubits encoding gates S .
- L data re-uploadings

Exponential growth accessible spectrum

Quantum Implicit Neural Representations



Quantum Visual Fields



Wang et al. Quantum Visual Fields with Neural Amplitude Encoding. NeurIPS, 2025.

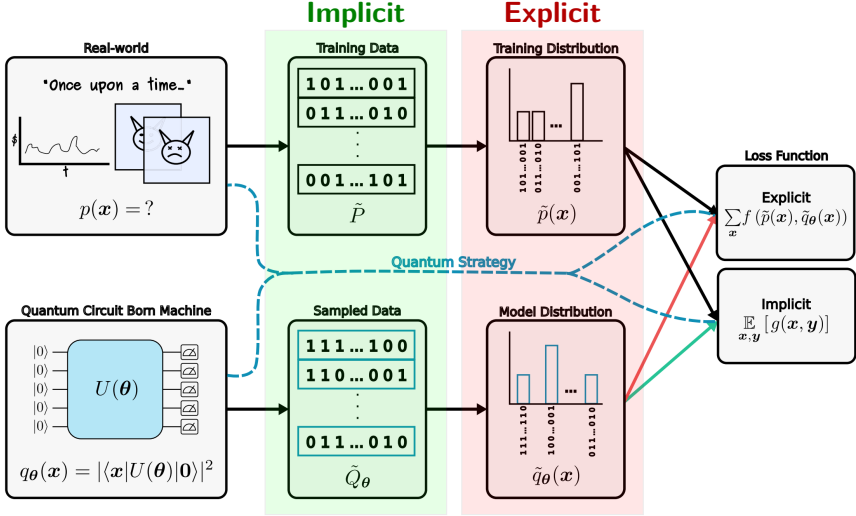
Park et al. DeepSDF: Learning Continuous Signed Distance Functions for Shape Representation. CVPR, 2019.

Quantum-enhanced Computer Vision: Going Beyond Classical Algorithms

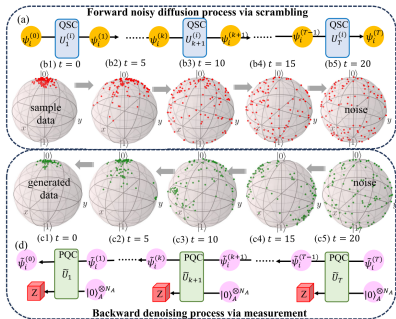
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Quantum Generative Models: Implicit vs. Explicit

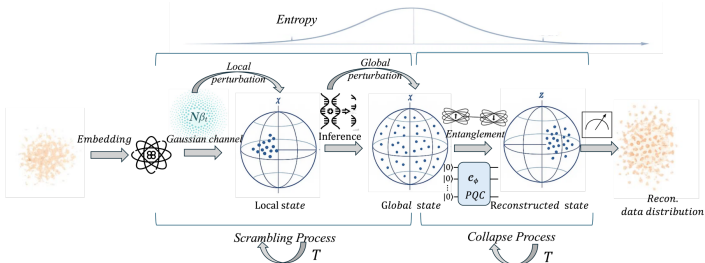


Quantum Explicit Generative Models with Diffusion



Quantum Denoising Diffusion Probabilistic Models (QDDPM):

- Hard to sample $x_T \sim p_T(x)$



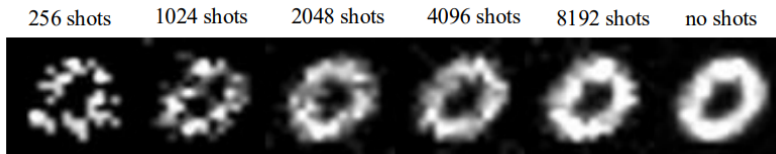
QDDPM with Gaussian Channels (QGen:)

- Gaussian noise for local perturbation
- Unitary scrambling for global perturbation

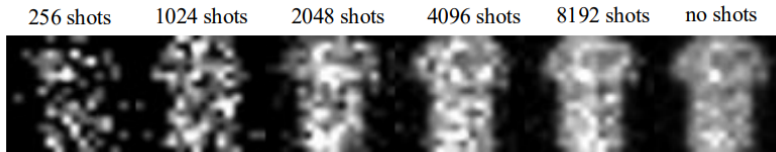
QDDPM: Zhang et al. Generative Quantum Machine Learning via Denoising Diffusion Probabilistic Models. Physical Review Letters 2024.

QGen: Li et al. A purely Quantum Generative Modeling through Unitary Scrambling and Collapse. ICLR 2026.

Quantum Explicit Generative Models



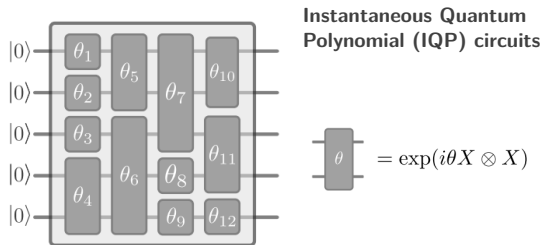
(a) MNIST



(b) FashionMNIST

Image generation with QGen.

Quantum Implicit Generative Models with IQPs



Training with MMD² loss

For a feature map ϕ and Gaussian kernel k :

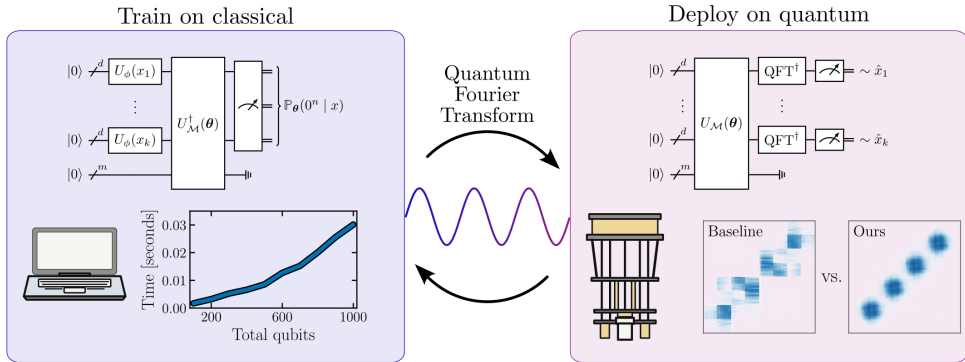
$$\begin{aligned} \text{MMD}^2(p, q_\theta) &= \|\mathbb{E}_{x \sim p}[\phi(x)] - \mathbb{E}_{y \sim q_\theta}[\phi(y)]\|^2 \\ &= \mathbb{E}_{x, y \sim p}[k(x, y)] \\ &\quad - 2\mathbb{E}_{x \sim p, y \sim q_\theta}[k(x, y)] \\ &\quad + \mathbb{E}_{x, y \sim q_\theta}[k(x, y)] \end{aligned}$$

It turns out that for IQP circuits (after some Hadamard sandwiches)

$$\begin{aligned} \text{MMD}^2(p, q_\theta) &= \mathbb{E}_{a \sim \mathcal{P}_\sigma(a)}[(\langle Z_a \rangle_p - \langle Z_a \rangle_{q_\theta})^2] \\ \langle Z_a \rangle_p &= \mathbb{E}_{x \sim p}[(-1)^{x \cdot a}] \\ \langle Z_a \rangle_{q_\theta} &= \mathbb{E}_{z \sim \mathcal{U}}[\cos(\sum_j \theta_j (-1)^{g_j \cdot z} (1 - (-1)^{g_j \cdot a}))] \end{aligned}$$

Train on Classical → Deploy on Quantum

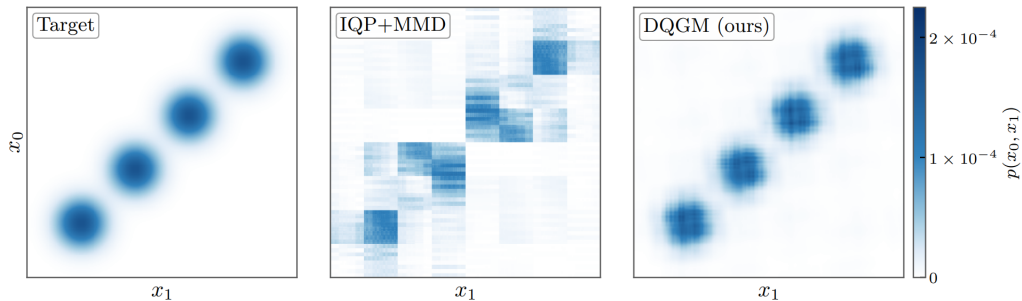
Quantum Implicit Generative Models with DQGM



Training with negative log-likelihood (NNL) loss

$$\text{NNL}(\theta) = -\mathbb{E}_{x \sim p}[\log P_{\theta}(0^n | x)], \quad P_{\theta}(0^n | x) = \mathbb{E}_{\omega \sim \mathcal{U}}[|\prod_{v=1}^n \frac{1}{2}(r_v(0; \omega) + r_v(1; \omega))|^2]$$

Quantum Implicit Generative Models

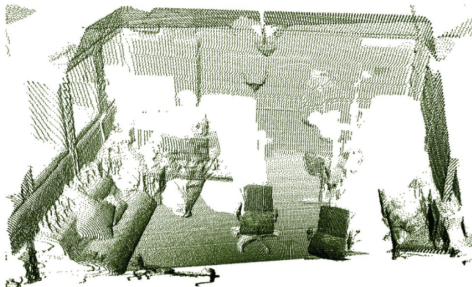


Open Challenges

Encoding, Training, Scaling

Encoding

How to **encode** high-dimensional vision data ...



... into a handful of qubits?

And how to **retrieve** classical signals from quantum states?

Animals dataset: <https://www.kaggle.com/datasets/alessiocorrado99/animals10>

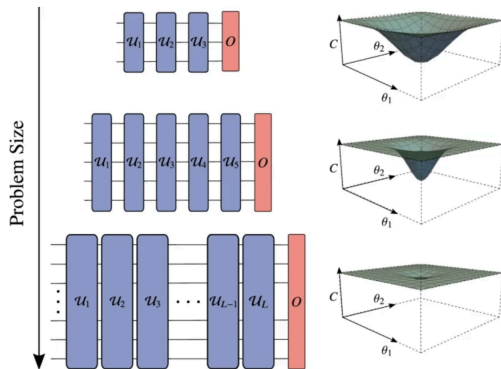
VARGA. 3D imaging and image processing – literaturereview. SCYR, 2016.

Training

Barren Plateau:

- Wide and deep PQCs
 $\Rightarrow$ higher expressivity ($\uparrow$).
- Wide and deep PQCs
 $\Rightarrow$ vanishing gradients ($\downarrow$):

“The gradient vanishes exponentially in the number of qubits n if the depth of the ansatz grows linearly with n .”



+ Sampling noise + parameter shift rules $\Rightarrow$ Prohibitive sampling overhead!

We need: Efficient gradient estimation methods, intrinsic quantum models (e.g., train classical $\rightarrow$ deploy quantum), etc...

Scaling

From proof-of-concept to realistic vision problems:

Data realism

- Beyond toy and synthetic datasets
- Real-world variability and complexity
- Large-scale

Hardware scaling

- Limited qubit count
- Connectivity constraints
- Decoherence limits computation time

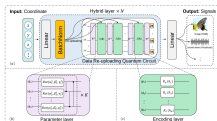
Execution cost

- Large number of circuit evaluations
- Probabilistic measurements
- Sampling overhead

Can quantum benefits survive at realistic scale?

We need: Theoretical guarantees or convincing demonstrations beyond small toy datasets, scalable and accessible hardware, etc...

Summary

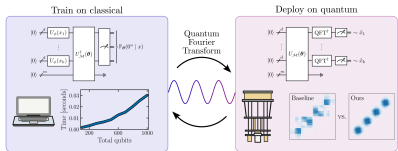


Adiabatic vs. Gate-based QC:

- AQC is a good heuristic for optimization.
Map CV problems to QUBOs first.
Prefer compact encodings, exploit sparsity.
- GQC offers flexibility in algorithm design.
Analytical is great but not obvious; training is an option.
Design around barren plateaus and sampling overhead.

Several emerging QeCV directions:

- Quantum computers are natural samplers.
- Quantum enables expressive spectral models.
- Exploit quantum–classical synergy.



Thanks! Questions?

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